

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement #06: Alumni Mentorship & Mock Interview Platform

Stakeholders / Actors: Student Mentee, Alumni Mentor, Calendar / Notification Service

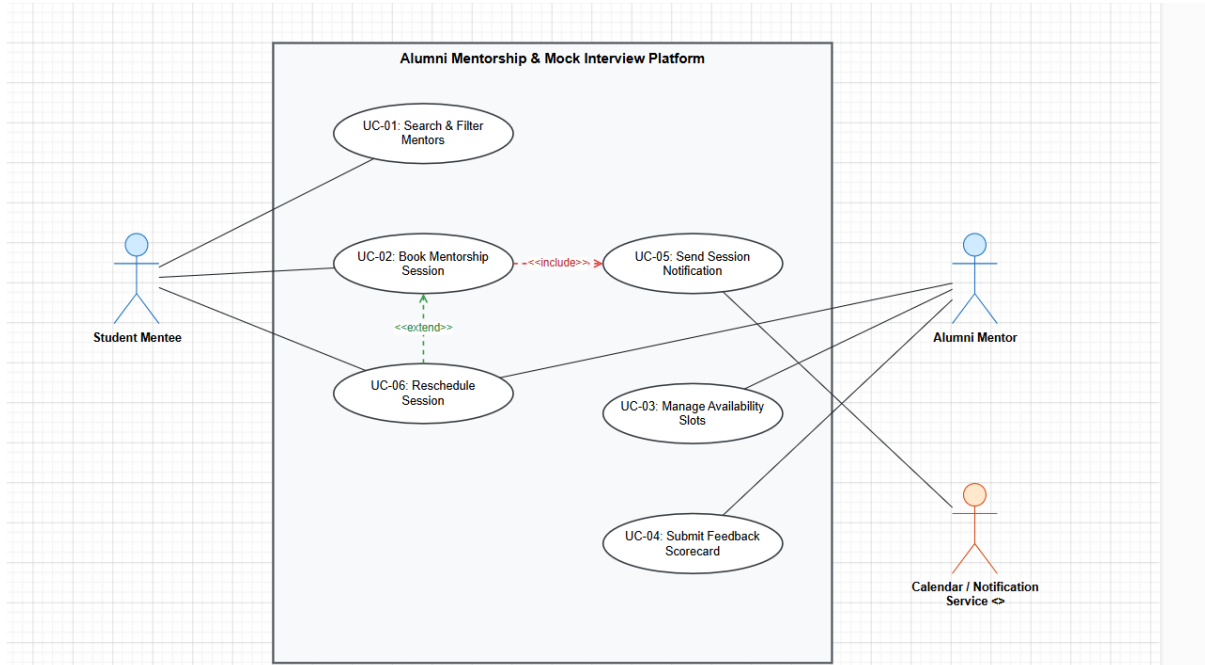
1. Requirements Specification Table

Req ID	Type	Description	Priority	ACCEPTANCE CRITERIA	Rationale (Short)
FR-001	Functional	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	High	Pass: Mentorship session is added to both calendars with a valid meeting link. Fail: Double-booking of mentor slots occurs.	Core functionality enabling intelligent pairing and scheduling.
FR-002	Functional	The system shall allow alumni mentors to set and update their available weekly time slots and domain expertise.	High	Pass: Updated availability reflects on the mentee booking screen immediately. Fail: Mentee can book during unavailable/unmarked hours.	Ensures accurate scheduling without scheduling conflicts.
FR-003	Functional	The system shall generate and send automated email and in-app notifications with calendar invites 24 hours and 1 hour	Medium	Pass: Both parties receive notification and calendar invite within the designated time window.	Reduces no-show rates for mock interviews.

		prior to the scheduled session.		Fail: Notification delivery fails or is delayed past the session start time.	
FR-004	Functional	The system shall enable alumni mentors to fill and submit a structured post-interview scorecard (ratings, rubrics, and feedback) for the mentee.	High	Pass: Scorecard is saved and accessible to the student mentee immediately after submission. Fail: Incomplete scorecards are submitted without required rubric fields.	Core value proposition providing actionable feedback to students.
FR-005	Functional	The system shall allow students to search and filter alumni mentors by company, years of experience, and specific mock interview topics (e.g., System Design, DSA).	Medium	Pass: Search results only show mentors matching all selected filter criteria. Fail: Unrelated mentors appear or search yields false negatives.	Helps students find specific mentors relevant to their immediate career goals.
NFR-001	NON-Functional	All user profile data and feedback scorecards shall adhere to privacy guidelines and support soft-deletion within 24 hours of request.	High	Pass: User profile and feedback data are soft-deleted from active views within 24 hours of request and cannot be retrieved by end-user APIs. Fail: Personal or scorecard data remains queryable/visible	Ensures compliance with data privacy regulations and user trust.

				past 24 hours post-deletion request.	
NFR-002	NON-Functional	The system shall maintain an end-to-end API response time of less than 2 seconds for recommendation and slot search under a concurrent load of 1,000 users.	High	Pass: At least 95% of search, recommendation, and slot retrieval API requests respond in ≤ 2 s under a concurrent load of 1,000 users. Fail: Response time exceeds 2 seconds or API fails under target concurrent load.	Guarantees platform responsiveness and smooth user experience during campus drive peaks.

2. UML Use-Case Diagram Specification



3. Use-Case Flow Specification

Use Case ID: UC-02

Use Case Name: Book Mentorship Session

Primary Actor: Student Mentee

Supporting Actors: Alumni Mentor, Calendar / Notification Service

- **Preconditions:**

1. The student is logged into the platform with an active, verified account.
2. The mentor has configured available open interview slots in their calendar.

- **Postconditions:**

1. The interview slot is marked as booked and removed from public availability.
2. A calendar invite containing a unique meeting link is generated and sent to both student and mentor.

Main Success Scenario:

1. Student searches for mentors by domain (e.g., Cloud Computing) and selects a mentor profile.
2. System retrieves and displays the mentor's calendar with available interview slots.
3. Student selects a desired open date and time slot.
4. Student enters mock interview focus areas and uploads their resume link.
5. Student clicks **Confirm Booking**.
6. System locks the slot to prevent simultaneous double-booking.
7. System triggers UC-05: Send Session Notification to schedule calendar invites with meeting links.
8. System displays a confirmation screen containing session details and meeting URL.
9. Use case ends successfully.

Alternate Flows:

- **6a. Slot Collision / Simultaneous Booking (Double-Booking Error):**

- **6a1.** System detects that another mentee completed booking for the identical slot milliseconds prior.
- **6a2.** System releases the temporary lock and displays an error message: *"This slot has just been reserved. Please choose an alternative slot."*
- **6a3.** System refreshes the mentor's availability calendar.

- **6a4.** Student selects a new time slot (Flow resumes from Step 3).
- **7a. External Calendar Service Timeout:**
 - **7a1.** System fails to generate the meeting link via the notification service API within 5 seconds.
 - **7a2.** System confirms the database booking reservation and queues the calendar link generation task in the background.
 - **7a3.** System notifies the mentee: *"Booking confirmed! Meeting link is being generated and will be emailed shortly."*